
Maintenance and Troubleshooting Guide

Includes:

Installation

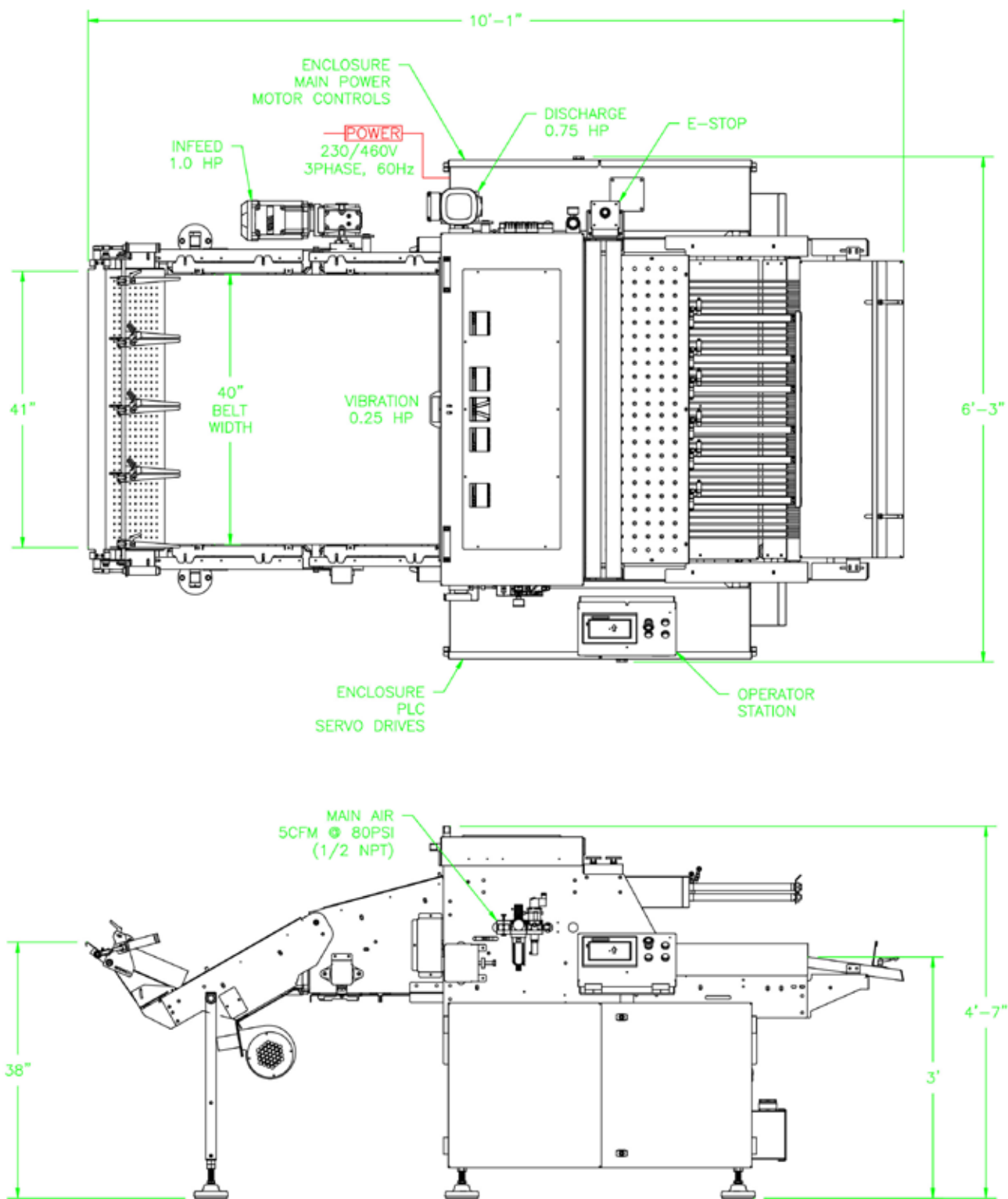
Components

Timers / Counts

Maintenance

Spare Parts

Troubleshooting



Installation

Remove the shipping crate, carefully.

Lift the Counter Stacker from the skid as instructed on page 16.

Caution: Use extreme care when lifting these machines.

Location

Position the Counter Stacker at the discharge end of your cooling conveyor.

Operating environment must be clean and dry.

The Counter Stacker requires 4ft (1.25m) clearance on all sides.

Infeed Conveyor

With the Counter Stacker at the discharge end of your cooling conveyor, adjust the infeed conveyor height.

- Loosen the jam nut at the leg base if necessary
Loosen the slide bolt on the support arm (on both sides).
- Position the conveyor approximately half an inch away from the cooler exit belt. It should be placed half an inch above the center line of the cooler exit belt.
- Turn the leg pad to lower or raise the conveyor to the desired height.

Always check belt tension after adjusting the infeed height. Any adjustment to the conveyor height will affect the belt length.

Leveling

Leveling of the Counter Stacker is recommended to preclude bumping, vibratory motion and interference. The leveling legs for each piece of equipment are located on the bottom of the frame.

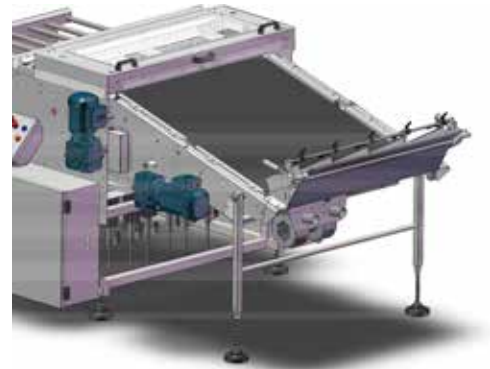
Adjust each leg pad so the wheels (option) at each end are approximately one quarter inch above the floor. The Counter Stacker must make solid contact at all points with the floor.

Arr-Tech recommends bolting the machines to the floor to maintain proper alignment at all times.

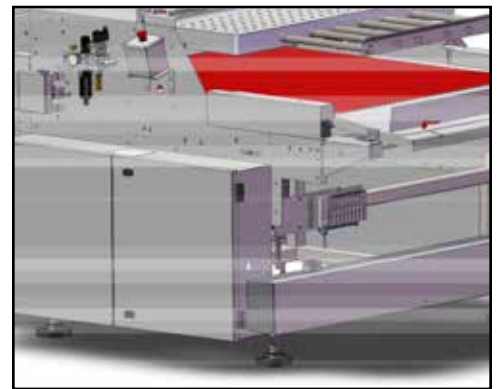
Offload Table

It is possible to adjust the position of the offload table (see figure 43-3).

If adjustments are made to this assembly insure that the gap between the discharge belts and the offload table is no more than 4mm.



*Figure 35-1
Infeed Conveyor*



*Figure 35-2
Leveling Leg*



*Figure 35-3
Offload Table Position*

Installation

Before Initial Use

Vacuum or blow any dirt or dust out of the unit.

Wipe all food contact surfaces with a damp cloth. Mist with a sanitizing spray, if desired.

Belts

Do NOT Attempt to clean the belts while they are in motion.

Push the EMERGENCY STOP button down to stop the belts.

Wipe the accessible portion of each belt with a damp cloth.

Release the EMERGENCY STOP button by turning it clockwise.

Push the SAFETY RESET button.

Push the CYCLE START button to advance the belts.

Repeat until the belts are clean.

Wipe all photo sensor lenses.

Blow out all hoses and air lines.

Wipe down the entire machine.

DO NOT Attempt To Wipe Down the Machine While it is Running.

Installation

Electrical

Ground Required.

This Counter Stacker operates on 230 volt, 3 Phase, 50 Hz, 11.8 amps electrical.

Allow 4ft (1.25m) workspace around the machine.

Electrical Hookup

Inspect the unit prior to electrical and air hookup.

Check for the following:

- possible shipping damage
- parts interference
- loose wires and air connections.

Electrical drops are recommended to be done from the ceiling.

Proper routing to avoid trip hazards are mandatory and must be properly marked.

Appropriate surge protection must be provided at the source.

Electrical supply is made in Enclosure A on the top of DS1001.

Minimum wire size is 12Awg or 3.31mm²

Electrical supply must penetrate the cabinet in either the left or right lower corner.

Supply Air

An air supply capable of delivering clean, dry air must be provided to the Counter Stacker to keep the equipment performing at optimal levels.

Air line requirements between the compressor and the counter stacker are as follows:

- 0-25 ft. ½" pipe
- 26-50 ft. ¾" pipe
- over 50 ft. 1" pipe.

Air Pressure

The Counter Stacker comes equipped with a filter and adjustable air regulator. Approximately 100 psi (6.9 bar) of pressure must be available. Air pressure to the machine must remain constant at a maximum of 70 psi (4.82 bar).

The air lines are color coded as follows:

White - Pressure
Blue - Open
Red - Closed
Black - Vacuum

Air Hookup

Air supply is made on the right side of the Counter Stacker, preferably from the top.

Air Requirements - 5cfm @ 80psi (5.6 bar)

Electrical and Air Service should be installed by a licensed contractor to the Counter Stacker. Quick Disconnect Ceiling Drops are recommended if permitted by local code.
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Installation

Remote Access option for machines that have Ethernet capability.

Remote access gives Arr-Tech Manufacturing the ability to access the PLC/Display or other device on the machine network via the internet. It provides an aid in troubleshooting or offering program enhancements without a service tech on site. The customer is only required to provide internet access to the machine.

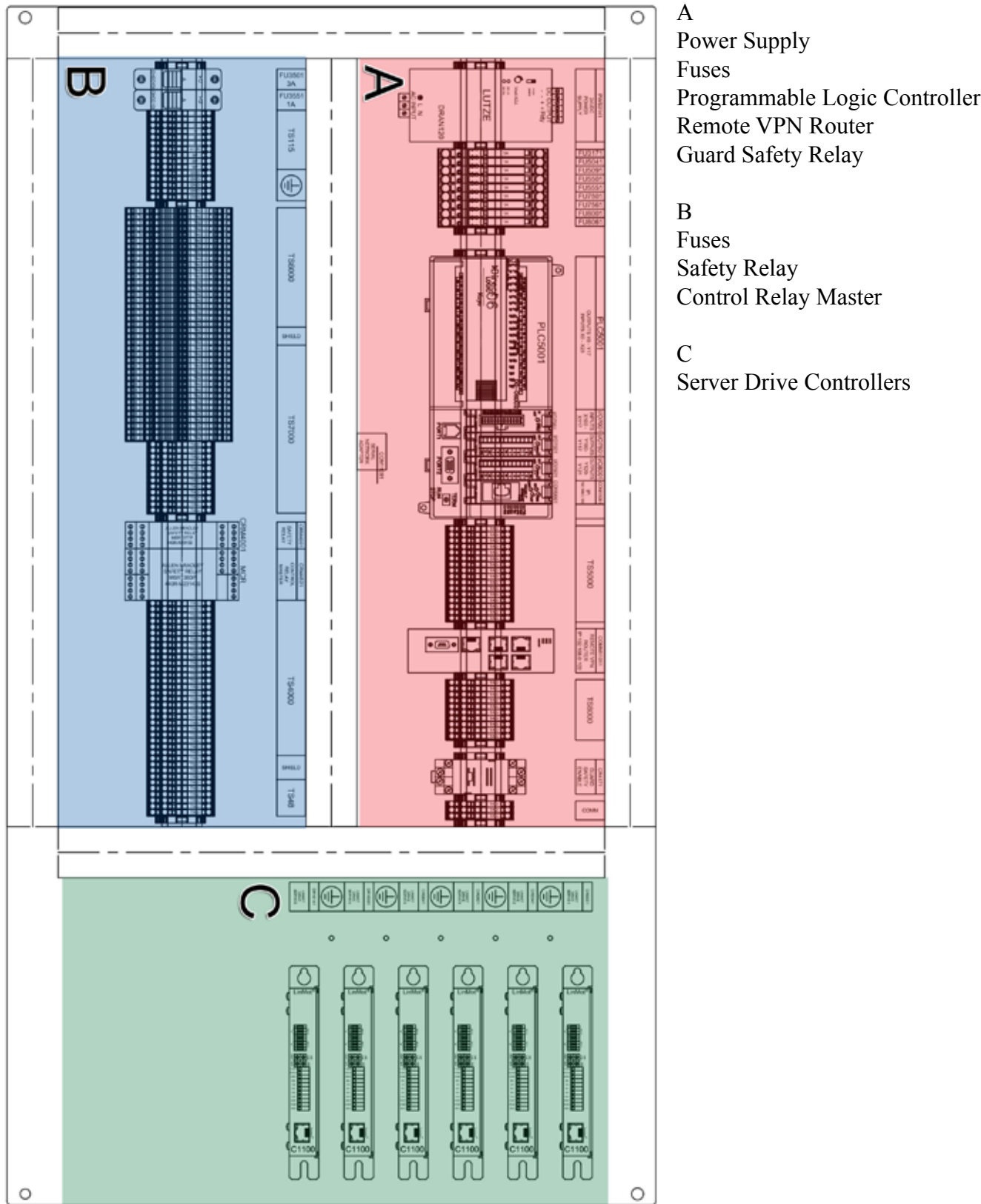
The device connects to the customer's network and requires setup to match the customer's plant network design. An Ethernet cable needs to be ran to the machine to provide the connection.

1. Is the network DHCP or Static IP addressing.
If it is Static IP we need an IP address supplied by the customer.
2. If no information is supplied The device will be setup with DHCP enabled and NO Static IP assigned.



Electrical and Air Service should be installed by a licensed contractor to the Counter Stacker. Quick Disconnect Ceiling Drops are recommended if permitted by local code.

Electrical Panels



Variable Speed Drive - Programming (Allen-Bradley Power Flex)

At right is a page From the PowerFlex Adjustable Frequency AC Drive Manual.

Drives

DR1501 Infeed Conveyor Drive Motor

DR1581 Discharge Conveyor Drive

Motor

DR1661 Stack Vibration Drive Motor

Editing parameters

When power is applied to the drive the last user selected display group parameter is displayed briefly before the current value.

1. Press ESC once to display the display group parameter that was shown on startup.
2. Press ESC again to enter the group menu, the group menu letter will flash.
3. Press the up or down arrow to scroll the group menu to "P".
4. Press "ENTER" to enter the group and the last digit of the last view parameter will flash
5. Press the up or down arrow until "P103" is displayed
6. Press "ENTER" to view the value. Press "ENTER" again to edit the parameter. The Program LED will illuminate if the parameter can be edited. (Press ESC to cancel)
7. Press the up or down arrow until the desired value is reached 5.5 for 220 volt machines and 2.5 for 440 volt machines. (Motor OL Current)
8. Press "ENTER" to save the change.
9. Press "ESC" to back out to the parameter list.

Repeat steps 5-9 for each parameter.

The following is an example of basic integral keypad and display functions. This example provides basic navigation instructions and illustrates how to program a parameter.

Step	Key(s)	Example Display
1. When power is applied, the last user-selected Basic Display Group parameter number is briefly displayed with flashing characters. The display then defaults to that parameter's current value. (Example shows the value of b001 [Output Freq] with the drive stopped.)		
2. Press Esc to display the Basic Display Group parameter number shown on power-up. The parameter number will flash.		
3. Press Esc to enter the parameter group list. The parameter group letter will flash.		
4. Press the Up Arrow or Down Arrow to scroll through the group list (b, P, t, C, L, d, A, f and Gx).	or	
5. Press Enter or Sel to enter a group. The right digit of the last viewed parameter in that group will flash.	or	
6. Press the Up Arrow or Down Arrow to scroll through the parameter list.	or	
7. Press Enter to view the value of the parameter. Or Press Esc to return to the parameter list.	or	
8. Press Enter or Sel to enter Program Mode and edit the value. The right digit will flash and the word Program on the LCD display will light up.	or	
9. Press the Up Arrow or Down Arrow to change the parameter value.	or	
10. If desired, press Sel to move from digit to digit or bit to bit. The digit or bit that you can change will flash.		
11. Press Esc to cancel a change and exit Program Mode. Or Press Enter to save a change and exit Program Mode. The digit will stop flashing and the word Program on the LCD display will turn off.	or	
12. Press Esc to return to the parameter list. Continue to press Esc to back out of the programming menu. If pressing Esc does not change the display, then b001 [Output Freq] is displayed. Press Enter or Sel to enter the group list again.		

Variable Speed Drive - Arr-Tech Settings (Allen-Bradley Power Flex)

Change to a different Parameter Group.

10. Press “ESC” again to back out to the group menu , The “P” will flash
11. Press the up or down arrow to change the “P” to the desired group designation (d, t, C, A).
12. Press “ENTER” to enter the group and the last digit of the last viewed parameter will flash.

13. Press “ESC” again to back out to the group menu , The “A” will flash
14. Press the up or down arrow to change the “A” to an “d”.
15. Press “ENTER” to enter the group and the last digit of the last viewed parameter will flash.
16. Press the up or down arrow until “d001 ” is displayed (Output Freq).
17. Press the stop button on the machine. Wait until the display goes blank on the display. Reapply power to the drive, it will start and you will be able to control the speed with the external potentiometer.

Parameter settings “as-shipped” can be found on the service CD on the program screen.

For more information about the operation of the Allen-Bradley PowerFlex Adjustable Frequency Drive refer to the User Manual which can be found on the Service CD or at www.rockwellautomation.com.

Air System: Air Preparation Unit

**Warning! Compressed Air!
Disconnect prior to servicing.**

Main Air Prep Unit

Manual Shut Off Valve (A)

The main air prep unit has a two-position gate valve that directs or shuts off high-pressure air to the machine. This shut off valve is lockable in the off position.

When the air flow switch (Gate) is set to off a lock can be installed to prevent unauthorized access.

Regulator (B)

A sight gauge shows the amount of air pressure in pounds per square inch (psi). For the machine to operate correctly this must read at a constant 80 psi.

The regulating hand knob adjusts the air pressure. It is located just above the sight gauge. Pulling up and turning the knob clockwise increases air pressure. Pushing the knob back down locks the adjustment in place.

Filter (C)

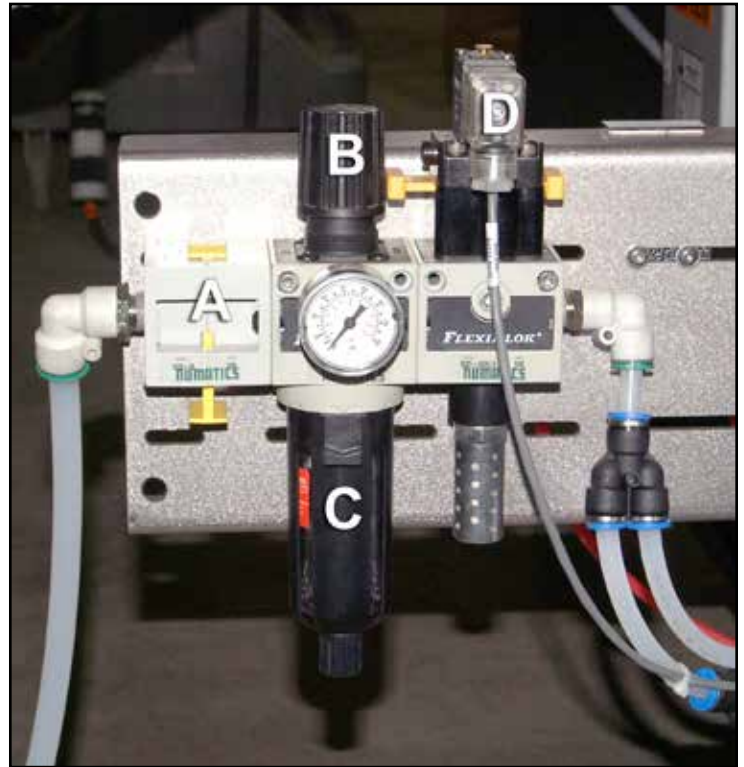
The filter assembly removes debris from the air system.

On the bottom of the filter assembly casing there is a screw-type pet cock for draining accumulated moisture.

Air Dump (D)

SOL4661

Drops air to the machine.



*Figure 43-1
Main Air Prep Unit*

Caution: Compressed air contains concentrated water vapor, oil, dirt particles, and other contaminations. If these contaminations are not removed they can damage the pneumatic equipment and controls. Clean Dry Air must be provided to the counter stacker to keep the equipment performing at optimal levels.

Air System: Solenoid Valves

A solenoid valve (figure 44-1) receives an electrical signal from the controller to shuttle a plunger back and forth. The plunger directs air from the air lines to the air-driven devices on the machine.

The manual override is a small blue screw-switch located on the front of the solenoid valve. Turn the switch using a screwdriver to override the electrical signal, which causes the plunger to open and direct air to the air-driven device.

Valves

5071	Diverter Gate
5081	Upper Interrupt Fork 1 In
5101	Upper Interrupt Fork 2 In
5111	Upper Interrupt Fork 3 In
5121	Upper Interrupt Fork 4 In
5131	Upper Interrupt Fork 5 In
5511	Upper Interrupt Fork 6 In
5561	Air Assist Nozzle 1 On
5571	Air Assist Nozzle 2 On
5581	Air Assist Nozzle 3 On
5591	Air Assist Nozzle 4 On
8011	Air Assist Nozzle 5 On
8021	Air Assist Nozzle 6 On
8031	Lower Interrupt Fork 1 In
8041	Lower Interrupt Fork 2 In
8051	Lower Interrupt Fork 3 In
8071	Lower Interrupt Fork 4 In
8081	Lower Interrupt Fork 5 In
8091	Lower Interrupt Fork 6 In



*Figure 44-1
Air Valves*



*Figure 44-2
Manual Override*